Predicting Content Similarity via Multimodal Modeling for Video-In-Video Advertising

Xue Song, Baohan Xu, and Yu-Gang Jiang

***Abstract*— Rapid development of mobile devices has led to explosive growth of videos and online platforms, which creates great demand for online advertising in videos. Existing advertising methods often aim to randomly select a time point as insertion position, which means that the video content is likely not related to the ad content, resulting in unsatisfactory user experience. While previous works have neglected to understand rich semantics as well as multimodal information in video advertising, in contrast to previous works, we present an innovative method for video-in-video advertising using multimodal modeling. First, different pre-trained models are used to extract multimodal representations. Then, through multimodal modeling, we learn the complementarity among different representations and obtain a unified video-level description. Finally, the unified representations of ads and videos are utilized to find the best matching result for each advertisement. Our method emphasizes the content similarity between ad and video, which would make the transition between video and ad more natural. Comprehensive experiments with both objective and subjective evaluations demonstrate the effectiveness and user-friendliness of our proposed video-in-video advertising framework.**

***Index Terms*— Video-in-video advertising, multimodal modeling, unified representation, content similarity.**

I. Introduction

IN RECENT years, the online video industry has developed rapidly. Users can easily watch nearly anything they desire from online video platforms, such as movies, TV shows, and variety shows. People across a wide range of ages and backgrounds are increasingly attracted by these platforms. Therefore, advertisements also arise in this newly emerged industry. The income from advertisements on YouTube in 2018 was over $10 billion [1]. Advertising revenue has become one of the main sources of revenue for online platforms. Thus, the platforms care deeply about how to allow users to watch ads for more than a certain duration of time in order to charge for additional advertising, which can benefit both platforms and advertisers.

In particular, the quality of advertisements is especially important to the choices by consumers of whether to watch an ad and/or buy a product. The content of an ad should tell a story or be situational, rather than fully commercialized, in order to reduce adverse ad effects, since most of the users will immediately close such commercialized ads as they appear. Regarding ad length, most long ads are edited to 15 or 30 seconds in order to successfully advertise for products. There are three main methods for inserting ads into videos. The most common method is to select a fixed time point to insert an ad, such as the fifteenth minute. Random insertion is another general method for ordinary advertisements. When a shot is going to change, an ad can be inserted without any relationship between the ad and the shot. These two methods can easily lead to an adverse effect, resulting in reducing the advertising revenue of the platform, or even affecting the brand image. There are some product placement advertisements [2] that have appeared recently which are indirectly played by the actors in the movies or TV shows. However, this is more expensive for production and is limited to be delivered in fixed content.

The above analyses motivate us to design a new method for video-in-video online advertising. From the perspective of users, they are more likely to watch ads in which they are interested: otherwise, they tend to close or ignore ads. From the platform point of view, they intend to increase the length of playing time of ads in order to acquire more advertising revenue. To tackle these challenges, we propose a novel video-in-video online advertising framework to balance the user experience and playing time of ads. Inspired by [3], [4], the authors advocate that an effective online advertising system should take both contextual relevance and reduced intrusiveness into consideration in terms of user experience. Our proposed framework focuses on making the content of the advertisement similar to the content of the video shot before the insertion point. We consider four aspects of the consistency of content: semantics, object, scene and sentiment. The semantics here refer to the topics of video shots and ads. For a better advertising effect, we also consider visual similarity of color, as well as aural similarity of audio. In this case, users are more likely to accept the appearance of the ad, and may not even notice the beginning of the ad, which invisibly increases the playing time of the ad and reduces the degree of intrusiveness caused by the ad. In order to utilize the complementary information between different modalities, we use a network to fuse different representations in order to form a unified representation, which can then be used as a high-level description for video content.

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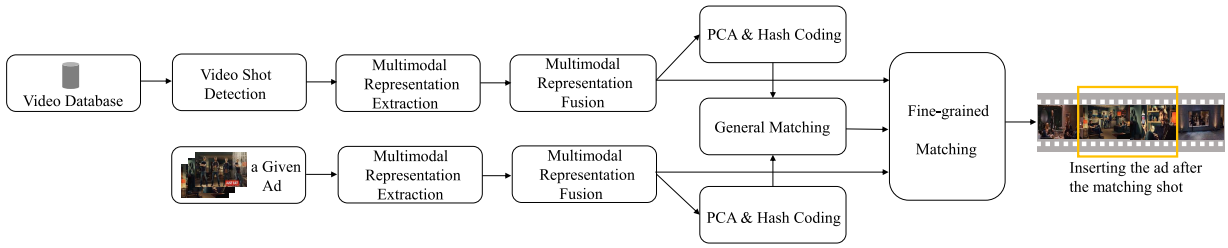


Fig. 1. Pipeline of our proposed video-in-video advertising system.

The pipeline is illustrated in Fig. 1. In particular, we first compute semantics, scene, sentiment, object and audio features with CNN models and compute color distribution histograms as color features. Then, a feature fusion network utilizes all of these features to derive a unified representation for predicting video semantics, which are video topics. The output before the final prediction layer is used as the unified representation for video content. Next, we calculate a unified feature for each video shot. For each unified feature, we apply principal component analysis (PCA) to reduce its dimension and then employ hash coding for computational efficiency. Finally, we use hash coding features for general matching to obtain a list of a smaller number of relevant shots. We use the unified representation of an ad to match the video shots in the list for fine-grained matching and insert the advertisement after the video shot with the best matching score. The main contributions of this work are summarized as follows:

1) We propose a multimodal modeling framework for video-in-video advertising, which includes semantics, scene, object, sentiment, audio and color. The high-level video features can not only capture the content of the ads and videos, but also help to select the best ad insertion point. The use of both general matching and fine-grained matching can leverage the time consumption and the matching accuracy.
2) We also explore the different multimodal fusion methods, including simple concatenation and fusion network. Our proposed fusion network is able to mine the interrelationships among different modalities.
3) Comprehensive experiments have been conducted to evaluate our proposed video-in-video advertising framework in terms of both objective and subjective aspects. Furthermore, some novel criteria are proposed for validating the user experience and ad performance, such as visual effects, aural effects and ad awareness, which are first investigated in our paper and may represent criteria for future studies.

II. Related Works

Extensive research has been conducted in the field of advertising. In the following, we first focus our discussion on domain-specific advertising and more general online advertising, such as Web advertising, which are more relevant to our paper. To further explore the similarity measurement of video-in-video advertising, existing visual matching methods are also discussed.

A. Advertising

Since the advent of video advertising, researchers have focused on monitoring and improving the effectiveness of advertising and the user experience. McDuff *et al.* [5] predicted ad liking from the perspective of analyzing facial responses of users, which can effectively portray user experience while watching an ad. As one of the most important categories of online advertising, the main goal of display advertising is to increase user response rates, such as click-through rates (CTR). Some researchers integrated different low-level visual features to predict CTR [6], [7].

As for the content analysis of videos, previous works have attempted to detect specific regions for insertion of commercial advertisements in sports videos [8], [9]. The proper location in [8] was identified by less-informative-region detection in the video. Wan *et al.* [9] detected and segmented goal-mouth appearances of soccer videos in real-time. However, these two systems have many limitations with respect to practical applications because they are aimed at specific domains and do not consider video content. Our video-in-video advertising system is designed for more general videos, rather than videos in specific domains.

To achieve general advertising success in recent years, Web advertising systems in the forms of pop-up ads, banner ads, floating ads and so on have become increasingly popular. However, the ways in which these ads are displayed causes user resentment and may lead to inferior ad effects. In order to overcome this problem, contextual advertising was proposed by [10], which dynamically shows ads related to Web pages. Image information was also considered in order to construct an image-based contextual relevance system for both text and image content within Web pages [11]. If the content of the ad was similar to what the user had searched for or browsed, the ad was more likely to be followed by the user [12]. Despite the advantages of contextual advertising, it also suffers from many limitations. For example, it requires publishers to rearrange the layout of Web pages and reserve space for inserting ads. The most popular contextual advertising system is Google's AdSense. It displays text and image ads that are related to the Web page content. Inspired by AdSense, Mei *et al.* [13] proposed a novel contextual in-video advertising system called VideoSense, which automatically detected relevant video ads and inserted them into a video at appropriate positions. Furthermore, VideoSense explored candidate ad lists according to surrounding text on the Web page. Later, the authors in [14] proposed a system called PageSense

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as a complement to existing contextual advertising, which supported not only semantically relevant advertising, but also style-consistent advertising. Although this system considered user experience to a greater extent, it did not consider the interests of advertisers. As mentioned in [14], how PageSense could simultaneously take both the viewers and advertisers into consideration still remained an open problem. In this paper, we take into account the users' feelings, in combination with the interests of the platforms and advertisers, to design an advertising system that benefits all.

To summarize, most related works have focused on domain-specific advertisement insertion and Web page advertising. There are fewer works focused on video-in-video advertising on online video platforms, which can only consider the contents of videos and ads. Videos have spread widely on the Internet and become a major information medium. As people increasingly demand video information, the application prospects in this area are becoming very broad. Some works consider object or clothing advertisements in videos. For example, Zhang *et al.* [15] focused on a comprehensive optimization model for object detection, which helped to minimize intrusiveness for video-in-video ads. Another work adopted several deep CNN models detecting human-body, human pose, face, and clothing [16]. These systems enable clothing-based video advertising but they can not process more general ads. Yadati *et al.* [17] proposed an in-stream video advertising strategy exploring the emotional aspects of video-in-video advertising, which was supported by empirical results from the marketing and consumer psychology literature. While the authors only took the emotion clue into consideration.

A universal advertising system is thus very necessary, and there is an urgent demand for this on platforms in which one problem is effectively addressed: how to select an insertion point for a given ad. In our proposed framework, the first consideration is that an ad should be placed at a point at which the video is progressing to the next shot, in order to avoid disturbing the continuity of the video and to reduce the interference by the ad on the original video content. Besides, this is a method which is more acceptable to users. Additionally, a given ad should be matched with each video shot in order to provide users with a consistently good experience, as well as to ensure better income for online platforms and advertisers.

B. Visual Matching

Many visual tasks such as classification, recognition, and tracking have taken advantage of visual matching, which aims to learn representations of videos or images to generate distance metrics to measure content similarity. Visual matching can be divided into two steps: feature description and metric learning. In order to cope with the large scale of data and limited computational capability, binary coding has been developed. The well-known coding strategy known as hash coding [18] is utilized in almost every computer vision field. Early researchers learned Euclidean distances, Hamming distances and Mahalanobis distances [19] for classification. Recently, with the development of deep neural networks and GPU technology, researchers have started to explore deep feature description and deep metric learning. The deep binary descriptor with multi-quantization was proposed [20] and outperformed most existing descriptors. Duan *et al.* [21] proposed a deep adversarial metric learning method to handle potential hard negative samples.

However, deep learning-based methods require more computational resources, such as GPU. This presents an obstacle for online deployment and application. To leverage the time consumption and accuracy, we perform two-stage feature post-processing, which includes general matching and fine-grained matching.

III. Methodology

In this section, we elaborate on the proposed video-in-video advertising framework. The complete framework is introduce at first place. Then we explain each part of the framework, including multimodal feature extraction, feature fusion and feature matching.

A. Overview

1) Preliminaries: For better understanding of our proposed system, the following terms are first clarified.

a) Ad: An ad is an advertisement video clip with length of 15 or 30 seconds, provided by advertisers, which can be inserted into a source video. In particular, the ads we used should tell a story, instead of directly advertising for products, which guarantees better user experience and proposes requirements for advertising production.

b) Source video: Source video, such as TV shows and movies, is usually played on online video platforms. In order to increase revenue, the platforms allow advertisers to insert ads into source videos at appropriate positions. Once an ad has been played for more than a certain period of time, the advertiser will pay the advertising fee to the platform.

c) Ad insertion point: The ad insertion position is a time point when the ad appears in the source video. At present, advertising usually selects a fixed time point to insert an ad, such as the fifteenth minute, which is a rigid method with less flexibility. It is appropriate to insert an ad at a point when the video shot is just about to change, which can preserve source video content against interference. Therefore, the candidate ad insertion points in our framework are shot switching points.

2) Framework: Fig. 1 shows the pipeline of our proposed video-in-video advertising system. First, a video database from the online video platform is built. Video shot detection methods are utilized to cut the video into shot-level clips. We then extract multimodal features and derive the unified representations for each video clip via our proposed fusion network. Our different modalities include semantics, object, scene, sentiment, color and audio. Moreover, each modality is necessary and plays an important role in representing video content. These unified features can be pre-computed and saved in a database for subsequent fine-grained matching. Besides, we first perform PCA for each unified feature, and then compute hash coding features to store in the database for

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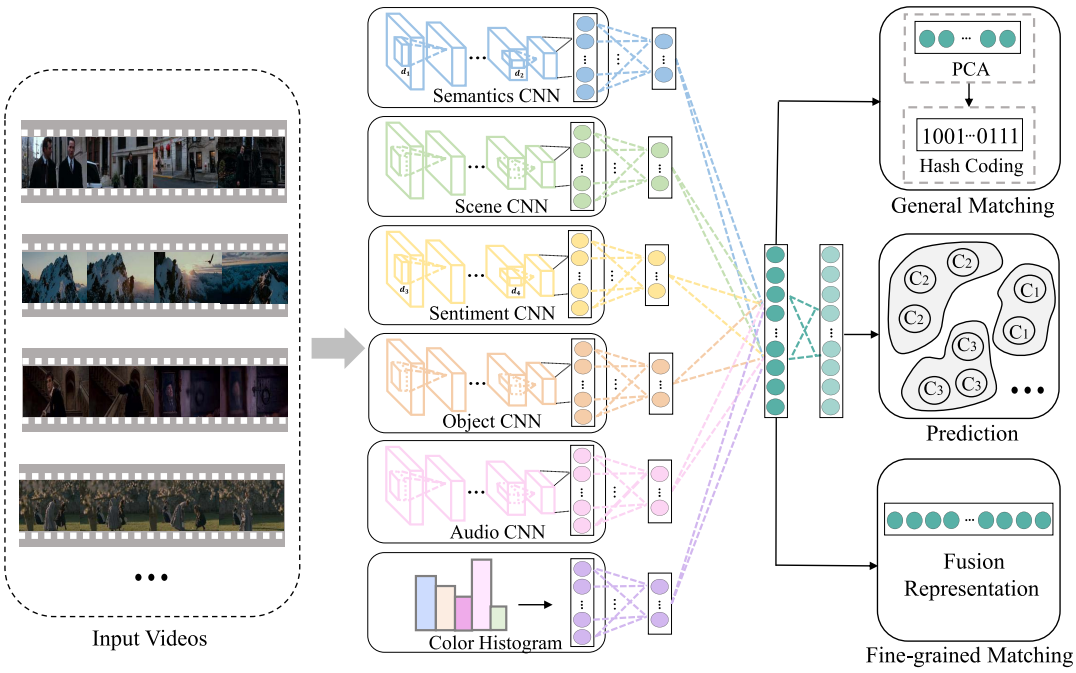


Fig. 2. Our proposed feature fusion network. Specifically, the semantic and sentiment CNN used 3D convolutional networks, where d indicates the depth dimension of the convolutional kernel. The scene, object and audio models all employed 2D CNN.

general matching. For a given ad, we calculate its unified representation and hash coding feature as described above. Next, we compute Hamming distance during general matching and compute the cosine distance between the representations of the ad and each shot in videos as similarity scores during fine-grained matching. Finally, the ad is inserted after the shot with the highest similarity score.

It is worth noticing that the method we propose imposes no limitation on ad content or video content. Thus, the framework can be widely used in online video platforms. We emphasize the importance of content matching between the ad and the video shot before the insertion point in reducing the sudden intrusiveness of the ad when the user is watching the video.

B. Shot Detection

The target tracking method [22] is used for our video shot detection. A pixel area in the center of the first video frame is regarded as a target to track. The video frame is determined to be a shot switching point when the target does not exist in it. This video frame will be used as the first frame for a new shot. The above steps are repeated until the end of the video.

However, in practice, this method often causes an over-segmentation phenomenon because of jittering. We observe that the over-segmented shots usually contain similar visual content, such as color distribution. To overcome this problem, color histograms is employed to remove some over-segmented points with similar global image color distributions.

C. Multimodal Feature Extraction

As shown in Fig. 2, we select six modalities to represent video content, each of which is indispensable.

1) Semantic Clue: The first clue we consider is semantics, namely topics. Naturally, the similarity of topics between ad and source video clips can be regarded as a part of the overall similarity. We use 38 topics defined in [23] for ads, such as cars, electronics, education, etc. Inspired by the promising results for action recognition, we take advantage of the I3D framework [24], which transforms C2D models into 3D convolution by inflating all of the filters and pooling kernels. The I3D network achieves state-of-the-art results for action recognition, demonstrating the strong power to represent video content. We use the pre-trained model on Kinetics [25] and fine-tune it with respect to our ad dataset for topic classification. Then the output of the last average pooling layer is used as the video-level semantic feature with 1024 dimensions.

2) Scene Clue: Scenes are highly related to the video content and ad content. For example, it would be more attractive for the video and the ad to both occur in a kitchen. More specifically, InceptionV3 is used as the backbone model pre-trained on ImageNet [26]. The Inception-style network exhibits superior performance with respect to the image classification problem. We then fine-tune the model on the Places dataset [27], which is a 10 million image database for scene recognition. The original image-level feature with 4096 dimensions is reduced to 512 dimensions after PCA processing for computational simplicity.

3) Sentiment Clue: Sentiment is also an important clue in our framework. Considering sentiment clues can help to avoid inappropriate situations such as inserting a funny ad within a sad video clip. We follow the same operations as semantic clue and use 30 sentiment categories defined in [23], like active, eager, and loving, to obtain 1024-dimensional video-level features.

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4) Object Clue: Furthermore, objects in videos often contain significant clues for video understanding. We use an open toolbox, MMDetection [28], and select the RPN [29] with the ResNet-50 backbone, which generates object bounding boxes of interest on each video frame and FPN [30]. This model applies ROIAlign [31] to extract the feature for each object proposal. RoIAlign generates a $7 \times 7 \times 256$ output feature for each object, which is then avg-pooled to $1 \times 1 \times 256$ dimensions. Finally, we obtain 256-dimensional features for each video frame.

5) Audio Clue: Existing works have proven that audio is another important clue in video analysis. More concretely, pre-processing used in [32] converts a.wav file to log-mel-scale spectrogram features as the input. An approximate mathematical conversion can be performed between the Mel scale and the linear frequency scale Hertz (Hz) [33]. The log-mel spectrogram is more in line with the hearing characteristics of the human ear since people's perception of sound is not linear, which is widely used in audio analysis tasks. We use the VGGish [32] associated with the dataset, called AudioSet [34], which is a VGG-like model and generates a 128-dimensional embedding for a given audio file as a video-level audio feature.

6) Color Clue: The color clue can not only benefit shot detection, but also contains substantial visual information. Color histogram is a major form of measuring color similarity between images. We calculate the color distribution histogram for each frame of three channels as a 256-dimensional image-level color feature.

Furthermore, we pre-calculate the features of the six modalities and store them for feature fusion. For the three video-level features, including semantics, sentiment and audio features, one video shot or ad will have a 1D feature. For the remaining image-level features, we uniformly sample video frames or ad frames to compute features and save all of them for the following process. Although the feature extracting methods are all state-of-the-art methods in different modalities, it is worth noting that our main contribution is to propose the multimodal modeling framework for predicting the content similarity of video-in-video advertising. Different networks for extracting multimodal features may slightly affect the feature representation, but would not change our conclusions with respect to the effectiveness of multimodal modeling in advertising.

D. Feature Fusion

The semantics, scene, sentiment, object, audio and color features characterize the same video content from different perspectives from video level to frame level. Exploring the inter-feature relationship between videos and ads can help us to understand multimodal information. To this end, we propose a feature fusion network to fully incorporate these relationships between different representations, rather than simply concatenating them, which ignores interactions between features.

Given a video shot, we first compute video-level semantics, sentiment, and audio features and frame-level object, scene, and color features. For frame-level features, we propose a weight-based combination method to generate video-level features. Taking scene features as an example, the other frame-level features are the same. We posit that the last frame of the video shot and the front frame of the ad play more important roles in the natural connection between these two parts in terms of content. Therefore, increasing weights is assigned to frames in the video shot and decreasing weights to frames in the ad instead of simply averaging them. Each modality can then be represented as a 1D feature for a video shot or an ad.

As shown in Fig. 2, semantics, scene, sentiment, object, audio, and color features are separately transformed into a higher space with one layer. We then use one layer to absorb all features to derive a unified representation. At last, the unified representations are used to generate final prediction scores for semantic classification: namely, topics. The unified representation we mentioned here is the final feature used for later matching, which contains multimodal information of video content.

More specifically, we represent the n-th training ad as a 7-tuple $(\mathbf{x}_n^s, \mathbf{x}_n^{sc}, \mathbf{x}_n^{sent}, \mathbf{x}_n^o, \mathbf{x}_n^a, \mathbf{x}_n^c, y_n)$, where $\mathbf{x}_n^s \in \mathbb{R}^{d_s}$ represents semantic features, $\mathbf{x}_n^{sc} \in \mathbb{R}^{d_{sc}}$ is for scene features, $\mathbf{x}_n^{sent} \in \mathbb{R}^{d_{sent}}$ represents sentiment features, $\mathbf{x}_n^o \in \mathbb{R}^{d_o}$ represents object features, $\mathbf{x}_n^a \in \mathbb{R}^{d_a}$ represents audio features, $\mathbf{x}_n^c \in \mathbb{R}^{d_c}$ represents color features and y_n is the corresponding groundtruth label. Now, we consider all training ads, and each ad has its six different features utilized as model input to train the fusion network. Let $g(\cdot)$ denote the nonlinear function approximated by the network. We use the following cross entropy loss to learn the optimal weights of the model:

$$\min_{w} \sum_{i=1}^{N} (-g(x_i)[y_i] + log(\sum_{j=0}^{C-1} exp(g(x_i)[j]))) \tag{1}$$

Here, x_i denotes the six input features for convenience, including $\mathbf{x}_i^s$, $\mathbf{x}_i^{sc}$, $\mathbf{x}_i^{sent}$, $\mathbf{x}_i^o$, $\mathbf{x}_i^a$, and $\mathbf{x}_i^c$. N denotes the number of ads in the training set and C denotes the number of topic labels. This fusion network is trained for producing better performance with respect to topic classification than the directly fine-tuned I3D model, which also proves that our fusion network can enable all-around understanding of video content.

The well-trained model is used to extract unified representation for each video shot in batches and store them in the database for later feature matching. Each representation has been processed to 1024 dimensions to balance the effectiveness and efficiency.

E. Feature Matching

As aforementioned, the unified representation and corresponding hash coding feature for each video shot are saved in the database. In practice, this is a convenient way for online video platforms to easily store and read the features for use. Furthermore, it saves a significant amount of time during the entire matching process. In general, there are millions of video shot features stored on the platforms. To speed up the matching, our process consists of two parts: general matching and fine-grained matching.

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For a given ad, we first calculate its six different features and derive the unified representation using our fusion network, and then compute the hash coding feature after PCA processing. The hash coding feature is considered as an 01 sequence for the unified representation. During general matching, the Hamming distances is computed between the hash coding features of the ad and each video shot. The smaller the Hamming distance between the two features, the higher the similarity between the two features. The Hamming distance is defined in Equation 2:

$$Hamming(\boldsymbol{x}, \boldsymbol{y}) = \sum_{i=1}^{N} sign((\boldsymbol{x} \oplus \boldsymbol{y})[i]), \tag{2}$$

where N is the dimension of $\boldsymbol{x}$, $\boldsymbol{x}$ and $\boldsymbol{y}$ separately represent hash coding features of the ad and the video shot and $sign(\cdot)$ denotes the element-wise sign function, which is defined as follows,

$$sign(x) = \begin{cases} 1, & x = 1 \\ 0, & x = 0 \end{cases} \tag{3}$$

Next, the video shots is sorted in ascending order according to the Hamming distance and utilize the first M shots (10000 in practice).

During fine-grained matching, cosine distances are computed between the ad and the representation of each video shot from the selected shots after general matching, which can be described in Equation 4.

$$cos(\boldsymbol{x}, \boldsymbol{y}) = \frac{\boldsymbol{x} \cdot \boldsymbol{y}}{||\boldsymbol{x}|| \times ||\boldsymbol{y}||}, \tag{4}$$

where $\boldsymbol{x}$ and $\boldsymbol{y}$ respectively represent fusion features from the ad and the video shot. We then take the cosine distance as the matching score and obtain a decreasing matching list according to score. The list records video shot information, including video name, the numbers of start and end frames, and corresponding matching score. Finally, the ad is inserted after the video shot ranked first in the matching list. If advertisers desire more ad insertion points, they can select video shots with higher scores from the list. The platforms can flexibly choose points based on the needs of advertisers and the limitations on the number of insertion points per video.

IV. Experiments

In this section, we first introduce the dataset and experimental settings. We then show both objective evaluation and subjective evaluation of our proposed advertising system, including user experience and ad performance.

A. Datasets

1) Ad Dataset: Our ad dataset consists of three parts in this work, with a total number of 3000 ads.

a) cvpr17: The first part we used is the video ad dataset publicly released in [23]. This dataset consists of 3477 ads in total from YouTube. However, at the time of our work, only 2535 ads were available. The average duration of the ads in this dataset is 50.8 ± 27.9 seconds. We also use the list of 38 topics and 30 sentiments defined in [23] as our semantic and sentiment label of single label classification. The experiments also prove that these two modalities make a great contribution to understanding video content.

b) aaai13: The second part includes the advertisement videos released in [35], with a total of 420 ads downloaded from YouTube. Jiang *et al.* collected ad videos in 14 topic categories; each category contains 30 videos. The average video duration of this dataset is 36 seconds. There are 4 ads in this dataset that duplicates with respect to cvpr17, and so we select 416 ads from [35] in total. To take full advantage of topic and sentiment models, we manually classify the 416 ads according to the labels in cvpr17 in terms of topics and sentiments instead of the 14 original categories.

c) online ads: The third part includes online ads crawled from online video platforms: there are 49 ads presenting beauty, houseware and other brands. Similarly, we classify them according to the topics and sentiments defined in [23].

2) Movie Dataset: We use the dataset in [36], called MovieQA, as the reference source videos. Although MovieQA aims to evaluate automatic story comprehension from both video and text, we only use the videos, since our general framework only focuses on video content. The video part consists of 6771 video clips obtained from 140 movies. The mean duration of the clips is 202.7 ± 216.2 seconds. Actually, we have downloaded 6718 movie clips. In order to ensure that a clip contains multiple shots, we remove the movie clips which are shorter than 20 seconds. Finally, the movie dataset for advertising consists of 4548 clips. After detecting and filtering the shots with fewer than 80 frames, we obtain 51958 movie shots, each of which is the basic unit of matching for a given ad.

B. Implementation Details

In this subsection, we introduce more details about our settings of single modality feature extraction model and fusion network. Regarding the dataset partitioning, we randomly divide 75% of the ads included in each semantic label as the training set, which consists of 2237 ads, and the rest as the testing set.

1) Video-Level Feature Extraction: The video-level features include **semantic features** and **sentiment features**. The ad dataset is used to train the I3D model for the two modalities separately, which is pre-trained on Kinetics, and take each ad as input for the model. During training, we randomly sample 64 frames, while 64 frames are uniformly sampled during testing. Furthermore, during training, we randomly select a 224 × 224 image crop to perform data augmentation, while during testing, we use the center crop from the video. We reference the above training settings in [24]. To fine-tune the model for semantic calssification, we first set the initial learning rate to 10^{-3} and decay it by a factor of 10 after 20 epochs, with a total of 30 epochs. Since the sentiment has less categories than semantic labels, the number of epochs is set to 15, which is

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enough for the model to converge. For feature extraction of an ad or a movie shot, we compute features of every 16 frames with no overlap as snippet-level features. The video-level representation is generated by averaging all of the snippet-level features.

2) Frame-Level Feature Extraction: The **scene features**, **object features**, and **color features** are regarded as frame-level features. A special frame selection strategy is adopted designing for matching ads and movie shots. Only frames from the first three seconds of the ad and the last three seconds of the movie shot is selected. The rest of the videos contribute less to the matching process. Then, we choose 1 frame every 3 frames to reduce computational cost.

As for object features, for each frame, we extract object features of object proposals and average them as frame-level object features.

As for color feature, each frame is first resized to the size of 256×256 with bilinear interpolation. We then compute color distribution histograms with 256 bins for each frame of the three channels and average the values to derive the frame-level feature.

3) Audio Feature Extraction: We first extract the audio of the first 3 seconds in the ad and the last 3 seconds in the movie shot. For an audio file, following the steps in [32], we compute log mel spectrogram features. These features are then framed into non-overlapping examples of 0.96 seconds, where each example covers 64 mel bands and 96 frames of 10 ms each. Next, these examples are treated as a batch of log mel spectrogram examples fed into the model. Finally, we average the output of this batch as a video-level audio feature.

4) Fusion Network Training: The input of our fusion network includes the pre-computed six different modality features, and the output includes the prediction scores for 38 topics. During training, the batch size is 4 and the learning rate is set to 10^{-4} for a total of 40 epochs. The change of learning rate is because that the fusion network is shallower than I3D meanwhile the features are more easy to converge or overfitting than images or videos as input. To avoid overfitting, we add a batch normalization layer [37], and dropout [38] is applied on the penultimate fully-connected layer with a ratio of 0.7. The loss function used in the fusion network is cross-entropy loss, which is the same as that of the I3D network.

C. Objective Evaluation

First, we conduct objective evaluations of our proposed fusion network. In our fusion network, six different modalities of features are used: semantics, scene, sentiment, object, audio and color. Empirically, each modality plays an important role in representing video content, which is essential for a better matching result of advertising. The objective evaluation focuses on topic classification, which represents the high-level concept of the whole video clip. This is also the most objective and fair way to compare the fusion network and each modality.

1) Comparison With the State-of-the-Art: To validate the effectiveness of the fusion network, we first compare it with three baseline methods. Firstly, we compare it with the method in [23], using C3D as the first baseline method. We also compared our proposed fusion network with some state-of-the-art video analysis methods including TSN [39] and I3D [24], which are the most commonly used methods for 2D convolution and 3D convolution, respectively. The compared methods show promising results for many video analysis tasks, such as action recognition and video classification. Besides, we extract the features of six modalities and simply concatenate them as the second baseline method. The evaluation criterion in the objective evaluation is the accuracy of topic classification.

TABLE I
PERFORMANCE OF STATE-OF-THE-ART METHODS FOR PREDICTING AD TOPICS

	Accuracy
C3D [23]	35.10%
TSN [39]	37.87%
I3D [24]	39.06%
Concatenating Features	39.19%
Fusion Network	**41.68**%

As shown in Table I, our fusion network with six modalities exhibits better performance than all baseline methods. Combining each modality is clearly useful for semantic classification. The results also demonstrate the complementarity of different modalities. The fusion network can help us learn the interrelationship among all of the modalities, as compared with simply concatenating features. Since semantics is closely related to video understanding, using the fusion network would also benefit the content representation in the following matching process.

2) Ablation Study:

a) Modality fusion: To further evaluate the impact of different modalities, we conduct the ablation study on different combinations of modalities. Table II reports the experimental results. The results demonstrate that different features can benefit topic classification in different ways. It is also worth noting that all of our fusion networks with different combinations of modalities outperform the baseline C3D and TSN methods. The complementarity of different modalities can help represent the ads and video shots more precisely with respect to both classification and matching.

It should be noted that the ad dataset we used in the experiments of Table I and Table II contains 38 topics, with unbalanced examples for each topic varying from 3 to 263. It is a significant improvement of 2.12% of multimodal fusion compared with only single semantic modality. Considering the difficulty of this task, we conclude that each modality makes its own contribution for topic classification from Table II.

b) Single modality: To take further investigation for each clue, the performance of single feature is also reported in Table III. It can be concluded that the semantic and sentiment features perform better than others and show their priorities. The performance of sentiment feature is second only to semantic feature because of the i3d network. While other features can still outperform the random result among 38 topics. In general, the semantics and sentiment features are

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TABLE II

PERFORMANCE OF DIFFERENT COMBINATIONS OF MODALITIES IN FUSION NETWORK FOR PREDICTING AD TOPICS

	Accuracy
Semantics	39.06%
Semantics + Scene	39.71%
Semantics + Scene + Sentiment	40.10%
Semantics + Scene + Sentiment + Object	40.24%
Semantics + Scene + Sentiment + Object + Audio	40.89%
Semantics + Scene + Sentiment + Object + Audio + Color	**41.68**%

TABLE III

PERFORMANCE OF DIFFERENT MODALITIES FOR PREDICTING AD TOPICS

	Accuracy
Semantics	**39.1**%
Sentiment	36.0%
Scene	19.0%
Object	17.3%
Audio	14.0%
Color	6.8%
Random	2.6%

the most important features containing the high-level concepts of the whole video clip. Besides, the audio and color features can be regarded as the auxiliary features to help improve the matching results. It is worth noting that the audio feature extraction model we used in our paper was pre-trained on approximately 20 billion examples from the 70M videos, which could achieve better performance for audio classification. These features all demonstrate the complementarity in modality fusion and matching.

D. Subjective Evaluation

User experience is one of the most important criteria of subjective evaluation in advertising. To improve user experience is the main goal of advertising systems. Furthermore, advertising performance is another essential aspect of real-world applications, especially for advertisers. Thus, we conduct user studies to evaluate our video-in-video advertising system with respect to both user experience and advertising performance.

1) User Experience: For comparison purposes, we create and adjust the following baselines of selecting ad insertion points: **Random selection,** which means that for a given ad, we randomly select a movie shot from the 51958-shot database and insert the ad after this movie shot. **Concatenating features,** where after separately computing the features of six modalities of a given ad, we simply concatenate them to form a new representation with 3200 dimensions. For each movie shot, we perform the same concatenating operation. We then match the ad with all movie shots, select the movie shot with the highest score as the final result, and insert the ad after this movie shot. **Fusing features,** where we first match the hash coding feature generated from the unified representation of the ad with those of all 51958 movie shots, reserve the top 10000 shots with the smallest Hamming distances, and then match the unified representation of the ad with those of these 10000 shots. Finally, we select the movie shot with the highest score as the final result. Our unified representation has only 1024 dimensions, which is more compact and efficient than the concatenated representation. **VideoSense [13],** where we follow the representative method to choose one type of similar feature. We use the video where the shot is located, which is selected by **Fusing features.** More specifically, we select a matching shot for the ad using the method proposed in [13]. For a fair comparison, our audio, color, and semantic features are used as their tempo, color, and concept features, respectively. As for the motion feature, the same method is applied as in [40].

50 ads were randomly selected from our ad dataset for evaluation. For each ad, we followed the four methods described above and determined the respective best matching movie shots. We found the movie clip where the matching shot was located, then inserted the ad after this shot. Eventually, we obtained 200 movie clips with ads inserted. More specifically, each one of these 200 clips had one ad inserted. Each method had 50 different advertising results. 9 evaluators were invited to participate in our experiments, which consisted of 5 female and 4 male participants with the age range from 18 to 30. Each person evaluated the advertising results of the same ads with respect to four different methods. To ensure a fair evaluation, the videos were shown in random order with no indication of the methods employed. Moreover, each advertising result was evaluated by three different people, and then the average score was calculated. Finally, the average score of 50 ads was calculated for each method. We evaluated four aspects of the user experience: ad relevance, ad intrusiveness, ad visual effects and ad aural effects. The visual and aural effects are first explored in our paper, since visual content and audio effects are both important factors of user experience.

a) Ad relevance: In order to evaluate the content relevance between each ad and the movie shot before the insertion point, each advertising result was judged by three participants on a (1-3) scale as follows.

a) Irrelevant (1): The ad is definitely unrelated to the movie shot before it in terms of content.
b) Somewhat Relevant (2): The ad is somewhat related to the movie shot before it in terms of content.
c) Relevant (3): The contents of the ad and the shot before it are essentially the same.

b) Ad intrusiveness: When watching a video, users usually feel averse to a sudden ad. However, existing advertising systems usually select a fixed ad insertion point. This approach does not consider the feelings of users at all, which may invisibly increase the degree of user resentment. In order to evaluate ad intrusiveness while watching a video, each advertising result was rated by three participants using integer values from 1 to 3.

a) Intrusive (1): The appearance of the ad is intrusive when watching a video.
b) Not Too Bad (2): The appearance of the ad is not overly intrusive.
c) Not Intrusive (3): This advertisement is appropriate and I am interested in watching it.

c) Ad visual effects: The overall color distributions of ads and video shots may also influence the user experience. A similar global color distribution can reduce the sense of

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TABLE IV
EVALUATIONS OF USER EXPERIENCE

	Relevance	Intrusiveness	Visual Effects	Aural Effects
Random Selection	1.42	1.61	1.41	1.39
VideoSense [13]	1.57	1.94	1.67	1.56
Concatenating Features	1.79	1.92	1.89	1.75
Fusing Features	**2.15**	**2.11**	**2.21**	**1.97**

visual abruptness. For the purpose of evaluating our proposed method with respect to ad visual effects, the participants were also asked to rate ads from 1 to 3 according to the following criteria.

a) Dissimilar (1): The ad is definitely dissimilar to the movie shot before it in terms of overall color distribution.
b) Somewhat Similar (2): The overall color distribution of the ad is somewhat similar to that of the preceding movie shot.
c) Similar (3): The overall color distribution of the ad is similar to that of the preceding movie shot.

d) Ad aural effects: When watching a video, an ad sometimes appears with abrupt audio that may disturb users. For example, if the ad contains music in the beginning, the system should choose a corresponding movie shot with music rather than talking in order to avoid interruption. We also create similar evaluation criteria to verify aural effects.

a) Dissimilar (1): The ad is definitely dissimilar to the movie shot before it in terms of aural sense.
b) Somewhat Similar (2): The aural effect of the ad is somewhat similar to that of the preceding movie shot.
c) Similar (3): The aural effect of the ad is similar to that of the preceding movie shot.

Table IV shows the evaluation results of user experience. A higher score indicates higher user satisfaction with respect to content relevance, lower intrusiveness, and more closely aligned visual and aural effects. As shown in the table, the method of fusing features exhibits very strong performance in terms of all four criteria, outperforming all of the compared methods. In particular, the relevance and intrusiveness scores demonstrate that features generated by fusing different modalities can improve user experience with more accurate content matching. The concatenating method performs slightly worse than the feature fusion, which indicates that the fusion network can fully utilize the complementarity of different modalities. The VideoSense method proposed in [13] performs worse than either fusing or concatenating features because of its limited single modality. The random selection method performs worse due to the sudden insertion without considering adjacent movie clips.

It is also worth noting that visual and aural sense are the two most intuitive feelings when watching a video, and that they are closely related to user experience. The results illustrate that color and audio features can be viewed as filling supplementary roles to help in decreasing ad abruptness. Color smoothing and audio smoothing are also utilized in practice to make users feel more comfortable.

The qualitative results of four methods with respect to ad insertion are shown in Fig. 3 with three different examples.

TABLE V
EVALUATIONS OF AD BRAND AWARENESS

	Number
Random Selection	7
VideoSense [13]	6
Concatenating Features	8
Fusing Features	8

As for the random selection method, it can be intuitively realized that the inserted ad is very abrupt with respect to both content and color distribution. As shown in the figure, the results of the latter three methods are significantly better than those of the first one. Moreover, from the perspective of the overall content, the video and ad employing feature fusion are closer than those employing feature concatenation and VideoSense [13] in all examples. This clearly shows that our fusion network can capture the content similarity of the entire video instead of just one perspective and we consider a comprehensive result for six modalities. This also demonstrates that ad relevance is closely associated with user experience.

2) Ad Performance: In this section, we consider the interests of the online video platforms and advertisers in order to design experiments to reflect the desired advertising performance.

a) Brand awareness: From the perspective of advertisers, the purpose of advertising is to cause users to remember the brand after watching the advertisement, and even to have the urge to buy the product(s). According to this purpose, we designed the brand awareness criterion to evaluate the performance of our advertising system. In particular, we randomly selected the advertising results of the same 15 ads with each method from the 50 results in the previous experiment. Another four participants were asked to watch 15 results of each method, respectively. We prepared a questionnaire with the logos of these 15 brands shown in the advertisements. Furthermore, the order of these 15 logos was inconsistent with respect to the order of viewing. What must be emphasized is that the participants are only asked to watch 15 videos, and so they do not know the intention of the experiments during the entire process of watching videos. After finishing watching fifteen movie clips with inserted ads, they were required to fill out the previously mentioned questionnaire and asked if they knew the 15 brands that they had just seen. Finally, we counted the number of brands which the participants still had impressions of as the experimental results.

As shown in Table V, participants remembered 8 of the 15 brands in the previously viewed videos, according to our methods, which is similar to the results obtained by the other

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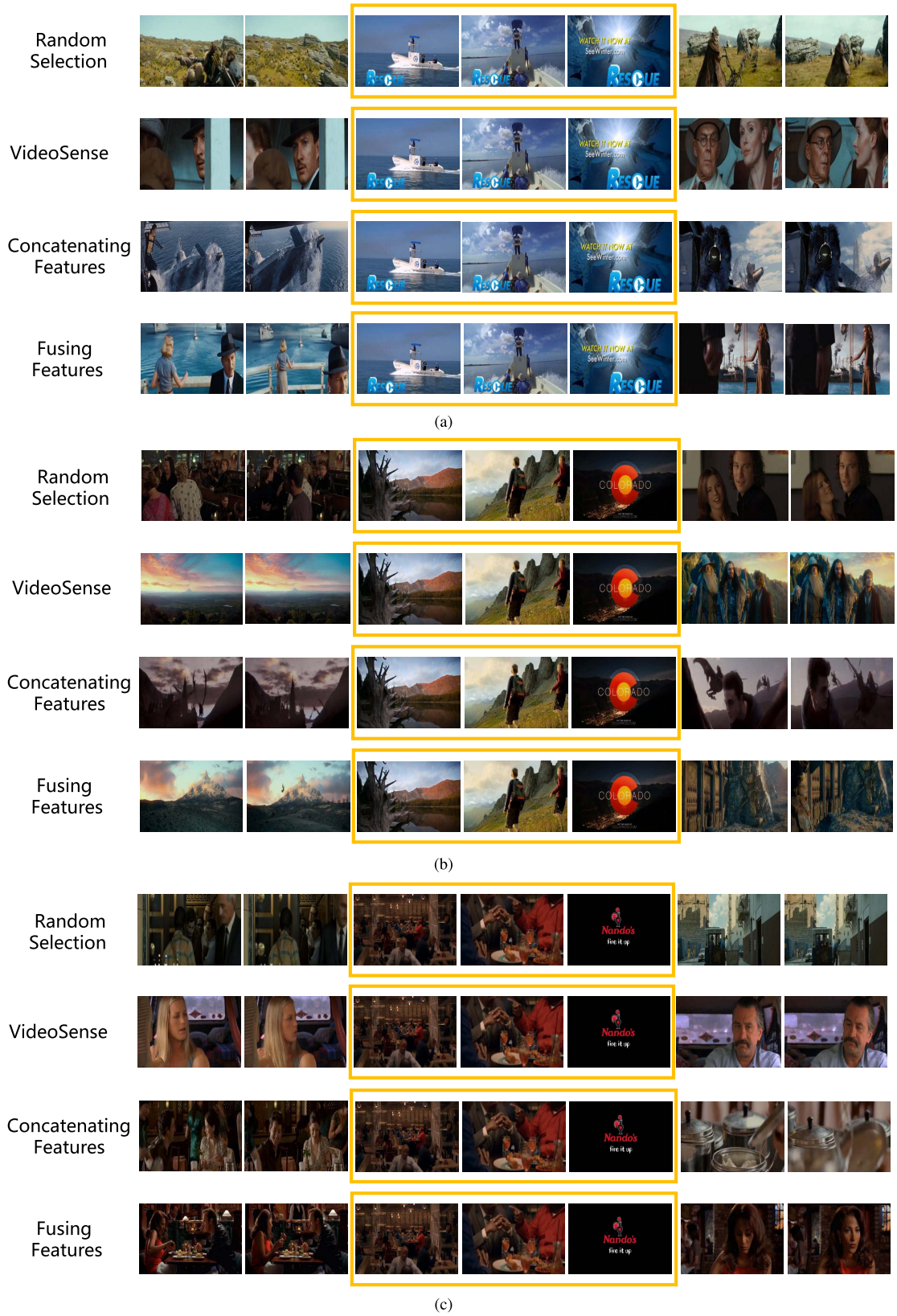


Fig. 3. Ad insertion examples with four different methods. The segment framed by yellow lines is the inserted ad. More vivid video examples can be viewed on: https://xxbbml.github.io/Video-In-Video-Ad/.

three methods. Intuitively, users may have deeper impressions when the images are suddenly changed. The results show that our proposed fusion method can lead to comparable brand impressions with better user experience. This may be because the relevant content makes the ads more interesting for users to watch, and therefore more memorable.

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TABLE VI
EVALUATIONS OF AD AWARENESS

	Average Value
Random Selection	1.47
VideoSense [13]	1.89
Concatenating Features	2.01
Fusing Features	2.48

b) Ad awareness: From the perspective of online video platforms, the playing time of ads is most important. When the advertising duration exceeds a fixed time, advertisers pay the corresponding advertising fee to the platforms. Based on the above considerations, we designed the following experiment for the first time. More concretely, we performed the same operation to select another 15 advertising results for each method and invited another 4 evaluators to participate in this experiment. In contrast with the Brand Awareness experiment, the participants were told that these videos were movie clips with an ad inserted. They were asked to record when they wanted to close the ad in this video after they noticed the ad. Furthermore, if they were not sure about the beginning of the ad, we would tell them the true position of the ad. Specifically, this experiment simulates the actual settings of an online video platform. For the purpose of evaluating our proposed method with respect to ad awareness, each advertising result was judged by one participant on a (1-4) scale as follows. Finally, the average score of 15 ads was calculated for each method.

a) At the beginning of the ad (1): The appearance of the ad is definitely abrupt, and the ad is more likely to be turned off right away.
b) Five seconds after the ad starts (2): The appearance of the ad is not overly sudden, and I want to watch it for a while.
c) Ten seconds after the ad starts (3): The appearance of the ad is acceptable, and I want to know more.
d) At the end of the ad (4): The ad which appeared here is attractive, and I unknowingly watched all of it.

The results of this experiment are reported in Table VI. With the score of 2.48, our proposed advertising system obtains a better grade than other methods. This illustrates that our method can increase the playing time of ads, which satisfies the requirement of online video platforms. It is also an interesting finding that only a few ads make people want to repeat viewing. We asked a participant who had seen a complete ad about his feelings. He said that the insertion of this ad was interesting, and that he was interested in the brand of the ad. We draw the conclusion that in addition to the importance of the location of the ad, user preference is also an important factor influencing the playing time of an ad. Besides, the quality of videos and ads would also affect the playing of an ad with respect to all of ad insertion methods. The largest contribution of our approach is to make the ad appear at the right time.

E. Computation Load

For an application system, the computation load is an important criterion to evaluate performance and practicability. The main computation load of our system includes two parts: feature extraction and feature matching. For a given ad, we can find the most closely matching video clip using our proposed method in less than three minutes. The specific analysis with respect to computation load is as follows.

TABLE VII
COMPUTATION LOAD OF FEATURE EXTRACTION FOR AN AD

	Extracting Time (seconds)	Params (M)
Semantic Feature	5.1	28
Sentiment Feature	5.1	28
Scene Feature	3.9	41
Object Feature	3.1	42
Audio Feature	1.6	136
Color Feature	0.2	-
Fusion Feature	0.1	3

TABLE VIII
COMPUTATION LOAD OF FEATURE EXTRACTION FOR AN AD

	Matching Time (seconds)
Fine-grained Matching	486.3
General Matching	51.9
Two-stage Matching	156.7

1) Feature Extraction: Since the features of online videos are calculated in advance and stored in the database, the time for feature extraction lies within the advertisement. Table VII lists the computation load of feature extraction for a given ad. It can be observed that the computational costs of semantic and sentiment features rank in the top two, which is caused by 3D convolution. In practice, we use 2 TITAN Xp GPUs to extract features in six modalities in parallel.

2) Feature Matching: Because of the large database of videos, we propose two-stage matching, which includes general matching and fine-grained matching. In Table VIII, we compared the computation loads of different settings. During fine-grained matching, the 1024-dimensional fusion feature of an ad is used for comparison with 51958 fusion features of video clips in the database and acquire the matching list, which requires 486.3 seconds. As for the general matching, it only requires approximately 1 ms per video clip. In our two-stage setting, we perform general matching to reserve the top 10000 related video clips, then follow fine-grained matching as described in the above section. The results show that our proposed two-stage matching greatly conserves time cost without reducing the accuracy of matching. Besides, adding more video clips into the database does not significantly increase the computational cost, due to the quick general matching. This two-stage matching method has more prospects in practical applications.

V. CONCLUSION

In this paper, we present a novel advertising system for online video platforms. More concretely, we emphasize the importance of matching between ad and video in terms of content similarity, which can reduce the intrusiveness of the appearance of the ad. In order to explore better representations to express video content, we learn the supplementary relationship between six different modalities, including semantics, scene, sentiment, object, audio and color. We then design a fusion network to model this collaborative relationship and

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derive a unified representation to characterize video content on a higher level.

Through an extensive set of objective and subjective experiments, we demonstrate that (1) our fusion network exploits the relationship between six modalities and enables better performance of topic classification for ads; (2) the six modalities we chose are indispensable for deriving a unified representation; (3) our advertising system performs well both in terms of user experience and ad performance. We evaluate the relative effectiveness of ad insertion strategies from three perspectives: those of users, advertisers and platforms. The experimental results provide strong evidence that our video-in-video advertising system achieves promising results.

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